Prateek Garg

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Academic Qualifications

Year	Degree	Institute
2017 - 2021	B.Tech in Computer Science	Indian Institute of Technology, Delhi

Scholastic Achievements

- **All India Rank 37** in *Graduate Aptitude Test in Engineering (GATE), 2026* (Mathematics), among **12 thousand** students
- **All India Rank 34** in *Graduate Aptitude Test in Engineering (GATE), 2025* (Mathematics), among **14 thousand** students
- Awarded **Outstanding Reviewer Award** for **Machine Learning Reproducibility Challenge, 2022** (*ReScience* Journal)
- **All India Rank 96** in *Joint Entrance Examination (Advanced), 2017* among **1.5 million** students
- **All India Rank 99** in *Joint Entrance Examination Main (B.Arch), 2017* among **150 thousand** students
- Received **Design Innovation Summer Award (2018),** competed against **hundreds of innovative projects**
- Awarded *Merit Certificate* for being in national **top 1%** in **Indian National Physics and Astronomy Olympiads, 2017**
- Awarded with **Kishore Vaigyanik Protsahan Yojana, 2016** Fellowship by **IISc Bangalore** securing **All India Rank 303**
- Completed **Vijyoshi National Science Camp, 2016** organised by **KVPY** with **IISc Bangalore** and ***Government of India***
- Awarded **Scholarship** for qualifying both stages of **National Talent Search Examination, 2015** by ***Government of India***

Professional Experience

- **Honda R&D Co. Ltd., AI Researcher | Honda Innovation Lab, Tokyo** (*Oct'21-Oct'23*)
  - **Advanced Driver Assistance System**: Built low-latency, low-cost smart DashCam; Presented POC to CXOs & board
  - **Cooperative Micro-Mobility**: Built robust *pedestrian trajectory prediction* framework & dynamic simulation environment Contributed to **path planning** and **robot vision** modules—*object detection*, *monocular depth*; wrote performance optimized *C++* code for real-time system. Acted as the main debugger, ensuring system stability and latency compliance
  - Participated in ICRA 2022: Reviewed papers, attended workshops, and engaged with the global research community
  - *Received special performance bonus in **2 out of 3 performance reviews** for exceptional results.*

Research Work and Internships

- **NLP Software Development Intern at LimeChat, India** (*Mar'21-Jun'21*)
  - Built new *tree* based *product quiz* feature for the chatbot and added capability to handle *multiple variants* of same product
  - Designed a new **data-management** framework to visualize training and finally improve the *intent* and *entity* classification
- **Research Intern at Alberta Machine Intelligence Institute, Edmonton, Canada** (*May'20-Jul'20*)
  - Conceptualized the effect of *state-similarities* on generalization in Monte-Carlo Tree Search using AlphaZero architecture
  - Built a bot on top of *KataGo*, state-of-the-art Go bot, using **Memory-Augmented MCTS** technique.
- **Research Intern at RIKEN Center for Advanced Intelligence Project, Tokyo** (*Nov'19-Dec'19*)
  - Built a Reinforcement Learning framework using **Deep Q-Learning** & **MCTS** for robots, integrating with ROS in Python
  - Designed toy problem for Car Racing AI setting using ROS, making the bot navigate its path through the maze optimally
- **Core Team Member at Robomuse 5.0, New Delhi** (*Discovery and Learn 1-2-3-4 Project*) (*Sept'18-Jan'20*)
  - Deployed **vision algorithms** for obtaining **Stereo-Image** & **extracting depth data** for SLAM purposes in mobile robot
  - Used standard navigation packages in **ROS** to develop complete back end for a fully **Autonomous Navigating Robot**
  - Designed a vision based **Landmark Recognition** technique for estimating global position state of the robot
  - Developed *Human Detection* and *Object Detection* modules in integration with ROS to add human safety features in bot
- **Research Intern at RIKEN Center for Advanced Intelligence Project, Tokyo** (*May'19-July'19*)
  - Performed an experimental study for the **traffic state prediction** during **Hiroshima disaster in June, 2018**
  - Prepared a comparative study between **RNNs** and Conventional Machine Learning Methods ***(SVMs, Random Forest)***
  - Conceptualized and merged the **effects of various features** on **Q-K curve** for critical nodes during the natural disaster
  - Compared interpretation of spatio-temporal features obtained from Machine Learning models with ***Traffic-Flow theory***
- **Project Associate at D-LIVE, Mahindra Autonomous Car Challenge, IIT Delhi** (*Feb'18-Aug'18*)
  - Received **Design Innovation Summer Award (DISA)** and sponsorship by *Institute R&D Dept., IIT Delhi*
  - **Improved odometry by 90%** at *35 kmph* using ZED Camera, GPS, IMU and obtained **correct loop closure results**
  - Worked on improving **ORB_SLAM2** to compute camera trajectory and sparse **3D reconstruction** for **visual odometry**
  - Implemented the **robot_localisation package** for accurate state approximation and integrating data obtained from GPS

  *Received a **Letter of Recommendation** for exemplary results and excellent contribution*

Academic and Technical Projects

- **Traffic Adaptability (B.Tech Thesis)**, **Prof. Rijurekha Sen**, Department of Computer Science, IIT Delhi (*Aug'20-Jan'21*)
  - Worked towards improving the adaptability of *traffic management models* based on **Meta Learning** techniques like **MAML**
  - Extended to develop a continual learning method for supporting the traffic model after deployment in an online fashion
- **PageRank**, **Prof. Rijurekha Sen**, Department of Computer Science, IIT Delhi. (*April'20-May'20*)
  - Developed an *open-source* implementation of the *Google's official* ***PageRank*** *algorithm* for large distributed systems
  - Used **MapReduce** algorithm with **MPI** in C++ for fast processing of *large graphs*, webpages. Tested upto **300,000** edges
- **Spine Classification & Segmentation**, **Prof. Prathosh AP**, Dept. of Electrical Engineering, IIT Delhi. (*April'20-May'20*)
  - Performed experimental study using Deep Learning techniques (**mainly CNNs**) to assess the spine structure from X-Rays
  - Used **U-Net** for segmentation and **Transfer Learning** from pre-trained model for pneumonia prediction for good results
- **AI Bots**, **Prof. Mausam**, Department of Computer Science, IIT Delhi. (*July'19-Sept'19*)
  - Gene String Mapping: Implemented ***Hill Climbing*** and ***Simulated Annealing*** search to find similarity between genes
  - The Game of Cannon: Made AI bot for the game by constructing ***MinMax Tree*** and implementing *Alpha-Beta Pruning*
  - Graph Subset Mapping: Formulated the problem as *constraint satisfaction problem* and further solved it by **SAT Solver**
  - Tic-Tac-Toe: Designed a computer player with various mastery levels using improved **Monte Carlo Tree Search** technique
- **Chat Application**, **Prof. Aaditeshwar Seth**, Department of Computer Science, IIT Delhi. (*Aug'19-Sept'19*)
  - Built a chat application similar to *WhatsApp* that allows user to do **encrypted chat**, which cannot be decrypted at server
  - Simulated on different network configurations by dynamically running for **multiple senders**, receivers on parallel threads

Independent Projects

- **Retentive Network**: Implemented and verified the seminal paper from scratch prior to official code base release.
- **Byte Level Subwords**: Tokenizer implementation and deep study from scratch in C++.
- **Neural Style Transfer**: Tried generating new artificial artwork using Transfer Learning.
- **RL Ensemble**: Implemented and visualised different RL algorithms on various OpenAI Gym environments
- **GANs**: Implemented and visualised the results of various GANs like DCGAN, Wasserstein GAN and Info-GAN, etc from scratch
- **Flipbook**: Developed special-language compiler for describing flipbooks & converting its description into printable PDF or GIF
- **Feature Hashing**: Implemented hashing technique described in the paper "*Feature Hashing for Large Scale Multitask Learning*"
- **OpenTrade**: Built a strategy backtesting & visualization tool for managing personal investments using 30 years data from *BSE*

Technical Skills and Relevant Courses

- **Skills:** C, C++, Python, Java, OCaml, Bash, Tensorflow, Keras, Pytorch, Sci-kit Learn, VHDL, ROS, OpenCV, OpenGL, MPI
- **Coursework:** Deep Learning, Pattern Recognition, Computer Vision*, Artificial Intelligence, Reinforcement Learning*, Learning for Cognitive Robot Intelligence, Programming Languages, Robotics Technology, Parallel Programming

Publications

- Short-term traffic prediction under non-recurrent congestion: A machine learning approach for traffic management during disaster *Makoto Chikaraishi,* ***Prateek Garg****, Varun Varghese, Kazuki Yoshizoe, Junji Urata, Yasuhiro Shiomi, Ryuki Watanabe (2019)*
- [Re:] Training Binary Neural Networks using the Bayesian Learning Rule ***Prateek Garg****, Lakshya Singhal, and Ashish Sardana (ReScience-C Journal)*

Patents

- ***Image recording device and medium****, Patent Application: US20260025484A1*
- ***Driver monitoring apparatus and medium****, Patent Application: US20260024358A1*
- ***Risk presentation apparatus and, risk presentation method****, Patent Application: US20260021824A1*
- ***Driver-assistance apparatus, driver-assistance method, and storage medium****, Patent Application: US20260024353A1*
- ***Drive recorder and control method thereof****, Patent Application: US20260015007A1*
- ***Driver-assistance apparatus, driver-assistance method, and storage medium****, Patent Application: US20260018053A1*

Past Supervisors

- **Jovin D'sa**
  Principal Engineer, Honda Research Institute, Tokyo, Japan
- **Prof. Martin Mueller**
  Canada CIFAR AI Chair at Amii, Department of Computing Science, University of Alberta, Canada
- **Dr. Kazuki Yoshizoe**
  Professor, Kyushu University, Japan